

Adaptive Personalized Federated Learning Framework for Privacy-Preserving Multilingual Next-Word Prediction

Kanishkaa Boopathi Raaj, Karthik Pranav Sivaraj and Gokul C

Department of Computer Science, Kongu Engineering College, Tamil Nadu, India

11 September 2026

ABSTRACT

On-device keyboards and text-input systems increasingly rely on next-word prediction models, yet training a single global language model on centrally pooled user text conflicts with user privacy and performs poorly for linguistically diverse populations such as Tamil-English code-mixed (Tanglish) speakers. This paper presents an Adaptive Personalized Federated Learning (APFL) framework that combines adaptive-rank Low-Rank Adaptation (LoRA), privacy-preserving local training using DP-SGD-style gradient clipping and Gaussian noise with FedProx regularization, and a Cross-Attention Gated Expert Router (CAGER) for personalized expert selection. The framework treats frozen per-client LoRA adapters as a mixture-of-experts pool and learns to combine them for a target client. Three CAGER variants are studied: a per-client router (v1), a shared multi-client router (v2), and a personalized router with expert-compatibility scoring and an own-expert prior (v3). The framework is evaluated on an 11-client, 6-corpus multilingual testbed spanning classical English, Tamil news, Tanglish, generic code-mixed text, Tamil movie reviews, and colloquial English. In the pilot configuration, the Adaptive PEFT baseline reaches 26.45% Top-1 and 42.68% Top-5 next-token accuracy averaged across clients. Personalized CAGER v3 matches Top-1 accuracy and improves Top-5 accuracy by 0.12 percentage points, while the naively shared CAGER v2 router collapses to near-zero accuracy. The ablation study shows that client-aware compatibility scoring and the own-expert prior are the key mechanisms that recover the routing performance lost in v2. The paper also discusses the limitations of pilot-scale training and directions for full-scale evaluation.

Keywords:

Federated Learning, Personalization, Parameter-Efficient Fine-Tuning, LoRA, Differential Privacy, FedProx, Mixture of Experts, Cross-Attention Routing, Next-Word Prediction, Multilingual NLP, Code-Mixed NLP

1. Introduction

Predictive text and next-word suggestion are core features of modern mobile keyboards, messaging applications, and assistive typing tools. The dominant approach to building such systems is to train a language model on large centrally collected text corpora. This approach raises two related problems. First, centralizing raw keystroke or message data creates serious privacy exposure, since informal text routinely contains personal, financial, and identifying information. Second, a single global model trained on generic corpora underperforms for users whose language use departs from the training distribution, for example, users who write in Tamil, Tamil transliterated into Latin script (Tanglish), or code-mixed Tamil-English, as well as users with distinct stylistic registers such as formal, colloquial, or archaic English.

Federated Learning (FL) addresses the privacy problem by keeping raw data on-device and exchanging only model updates, but standard FL algorithms such as Federated Averag-

ing (FedAvg) still converge toward a single shared model. Under statistical heterogeneity typical of real user populations—including non-identically distributed vocabularies, scripts, and writing styles—a single global model is a compromise that serves the average client well and atypical clients poorly. This motivates personalized federated learning, in which each client retains a lightweight, locally adapted component while benefiting from federated structure and shared computation.

This paper proposes an Adaptive Personalized Federated Learning (APFL) framework for privacy-preserving next-word prediction that combines three complementary ideas. First, personalization is achieved through Low-Rank Adaptation (LoRA), a parameter-efficient fine-tuning technique that freezes a shared pretrained language model and learns a small low-rank update per client. The LoRA rank is selected adaptively from a personalization score computed from the client’s available data volume. Data-rich clients receive higher model capacity, while data-poor clients receive smaller and more regularized adapters.

Second, local training is made privacy-aware through per-step gradient clipping and Gaussian noise injection in the style of DP-SGD, combined with a FedProx proximal penalty that anchors each client’s adapter to the shared initialization and stabilizes learning under non-IID conditions.

Third, the framework introduces a Cross-Attention Gated Expert Router (CAGER) that treats each client’s frozen LoRA adapter as an expert and learns, for a given input and target client, a sparse top- K combination of experts. Three CAGER variants are developed and compared: a per-client router (CAGER v1), a router shared across all clients (CAGER v2), and a personalized router that adds expert-compatibility scoring and an own-expert prior (CAGER v3).

The framework is instantiated on DistilGPT2 as the shared backbone and evaluated on an 11-client federated testbed drawn from six multilingual and code-mixed text corpora: Shakespearean English, Tamil news, Tanglish, generic code-mixed text, Tamil movie reviews, and colloquial conversational English. The combination deliberately stresses personalization through differences in script, register, and language-mixing behavior.

The main contributions are:

- An adaptive-rank PEFT scheme in which each federated client’s LoRA rank is selected from a data-driven personalization score using rank-tiered thresholds and gradient masking on a single physical adapter.
- A privacy-preserving local training procedure combining DP-SGD-style gradient clipping and Gaussian noise with FedProx regularization.
- A family of three Cross-Attention Gated Expert Routers (CAGER v1, v2, and v3) that reuse frozen client LoRA adapters as a mixture-of-experts pool.
- An 11-client, 6-corpus multilingual and code-mixed federated next-word-prediction testbed spanning English, Tamil, Tanglish, and code-mixed registers.
- A pilot evaluation with per-client analysis and ablation of client-awareness, cross-attention, expert compatibility, and the own-expert prior.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the proposed architecture. Section 4 presents the experimental setup. Section 5 reports results and discussion. Section 6 presents the ablation study. Section 7 discusses limitations. Section 8 concludes the paper and outlines future work.

2. Related Work

2.1. Federated Learning and Personalization

Federated Averaging established the standard paradigm for training a shared model across decentralized clients without transmitting raw data. Subsequent work showed that FedAvg can degrade under statistical heterogeneity, motivating person-

alized federated learning. FedProx introduces a proximal term that penalizes local model drift from the global model, improving stability and convergence under non-IID data.

Other personalization strategies include per-client fine-tuning, model-agnostic meta-learning approaches such as PerFedAvg, and multi-task formulations such as Ditto. Federated learning for mobile keyboard prediction provides an important applied precedent for the privacy motivation of this work.

2.2. Parameter-Efficient Fine-Tuning

Fine-tuning all parameters of a pretrained language model per client is expensive in federated settings because of local computation and update size. Parameter-efficient fine-tuning (PEFT) freezes the pretrained backbone and trains a small number of additional parameters.

Adapter modules insert bottleneck layers, while LoRA learns a low-rank decomposition of the weight update. LoRA is particularly attractive for federated personalization because its rank directly trades representational capacity against parameter count, communication cost, and overfitting risk. This property is exploited by making the rank a client-specific adaptive hyperparameter.

2.3. Differential Privacy in Federated Learning

Although FL avoids transmitting raw data, model updates may still leak information about local training examples. Differentially private stochastic gradient descent (DP-SGD) limits this leakage by clipping gradients and adding calibrated Gaussian noise.

DP-FedAvg extends the idea to federated learning by clipping and perturbing client updates before aggregation. The present framework applies clipping and noise to local LoRA gradient steps, combining privacy protection with a small trainable parameter budget.

2.4. Mixture-of-Experts and Routing

Mixture-of-Experts (MoE) architectures route inputs through sparse subsets of specialized subnetworks. Sparsely gated MoE and Switch Transformer models demonstrate the effectiveness of top- K routing.

This work adapts the MoE concept to federated personalization: each independently trained and frozen client LoRA adapter is treated as an expert, while a lightweight cross-attention router learns to combine the experts without modifying them.

2.5. Multilingual and Code-Mixed Language Modeling

Multilingual users frequently mix languages and scripts within a single utterance. Tamil-English code-mixing (Tanglish) is a representative example of this phenomenon.

Predictive-text systems trained primarily on monolingual English corpora can generalize poorly to such inputs. The proposed evaluation therefore spans classical English, native-script Tamil, Tenglish, generic code-mixed text, Tamil movie reviews, and colloquial English.

3. Proposed Framework

3.1. System Overview

The framework is organized as an eleven-module pipeline:

- (i) multilingual data loading,
- (ii) text preprocessing and cleaning,
- (iii) DistilGPT2 tokenization for causal language modeling,
- (iv) partitioning of tokenized data into federated clients,
- (v) adaptive PEFT model construction,
- (vi) client personalization profiling,
- (vii) privacy-preserving adaptive client training,
- (viii) CAGER v1, v2, and v3 router training and evaluation,
- (ix) comparative and ablation analysis,
- (x) final results aggregation and reporting, and
- (xi) an interactive next-word-prediction demonstration with automatic client-profile detection.

Each stage after the initial federated client training operates on frozen artifacts produced by the previous stage.

3.2. Backbone Model

The shared backbone is DistilGPT2, a distilled six-layer causal transformer language model with hidden size 768.

The base model is frozen in its entirety; all client-specific learning is confined to LoRA modules attached to its linear projection layers.

3.3. Adaptive-Rank Low-Rank Adaptation

Each client is assigned a LoRA adapter with a maximum physical rank of 16, $\alpha = 32$, and dropout = 0.1, attached to attention and feed-forward projection layers.

A single maximum-rank adapter is instantiated and only the first $r \leq 16$ rank dimensions are activated for a client.

Unused rows and columns of the LoRA matrices are zeroed, gradient hooks mask inactive rank dimensions, and the LoRA scaling factor is rescaled to α/r .

This keeps one adapter shape across all clients while providing different effective capacities.

For maximum active rank 16, the adapter contains 811,008 trainable parameters against 81,912,576 frozen backbone parameters, corresponding to approximately 0.98% of the total 82,723,584 parameters.

3.4. Personalization Profiling and Rank Allocation

The personalization score for client c is computed as

$$s_c = \frac{n_c}{\max_c(n_c)}, \quad (1)$$

where n_c is the number of local training samples.

The active LoRA rank is selected using fixed tiers:

$$r_c = \begin{cases} 16, & s_c \geq 0.70, \\ 12, & 0.40 \leq s_c < 0.70, \\ 8, & 0.15 \leq s_c < 0.40, \\ 4, & s_c < 0.15. \end{cases} \quad (2)$$

The assumption is that clients with more local data can support higher capacity, while data-poor clients benefit from stronger regularization.

3.5. Privacy-Preserving Local Training

Each client trains its LoRA adapter locally using a causal language-modeling loss augmented with a FedProx proximal term:

$$\mathcal{L}_{\text{prox}} = \frac{\mu}{2} \|\theta - \theta_{\text{global}}\|^2, \quad (3)$$

where $\mu = 0.01$.

When differential privacy is enabled, the trainer clips trainable-parameter gradients to $C = 1.0$ and adds Gaussian noise with standard deviation

$$\sigma = \frac{2Cz}{B}, \quad (4)$$

where $z = 0.1$ is the configured noise multiplier and B is the per-device training batch size.

The sequence is clip, noise, and update. FedProx stabilizes optimization under heterogeneous data, while gradient clipping and Gaussian noise provide DP-style protection against gradient leakage.

3.6. Cross-Attention Gated Expert Router

After the 11 client LoRA adapters are trained and frozen, they form a fixed pool of experts.

A client-context representation attends over the expert representations using scaled dot-product cross-attention.

A gating network with hidden size 256 produces probabilities over the 11 experts.

The distribution is sparsified to the top- $K = 3$ experts using a softmax temperature of 1.0.

Only router parameters are trained during this stage; the base model and client adapters remain frozen.

Table 1. Dataset composition and federated client assignment.

Corpus	Language / Domain	Text Field	Full Size	Clients
Shakespeare	Classical English	PlayerLine	108,741	3
Tamil News	Tamil (native script)	NewsInTamil	14,521	2
Tanglish	Tamil-English	text	—	2
CodeMix	Generic code-mixed	text	—	2
Movie Reviews	Tamil movie reviews	ReviewInTamil	—	1
Colloquial	Conversational English	Colloquial_Text	—	1

3.6..1. CAGER v1 – Per-Client Router

CAGER v1 trains an independent router for every client using only that client’s held-out calibration batches.

It serves as a sanity check for cross-attention routing over frozen experts.

3.6..2. CAGER v2 – Shared Multi-Client Router

CAGER v2 trains a single router shared across all 11 clients using a small per-client calibration set.

In the pilot configuration, 10 samples per client are used.

The configuration exhibits severe degradation because the shared router lacks a client-specific compatibility signal.

3.6..3. CAGER v3 – Personalized Router

CAGER v3 retains the shared cross-attention architecture and adds an expert-compatibility term with weight 1.0 and an own-expert bias with weight 1.5.

A personalization weight of 0.5 balances these terms against the attention-derived gate logits.

These additions encourage the requesting client’s own expert to be selected unless another expert is demonstrably more compatible.

3.7. Evaluation Protocol

Every configuration—Base DistilGPT2, Adaptive PEFT, CAGER v1, CAGER v2, and CAGER v3—is evaluated per client using causal language-modeling loss, perplexity, Top-1 next-token accuracy, and Top-5 next-token accuracy.

CAGER experiments additionally record expert gate distributions, top- K expert selection frequencies, and own-expert selection rates.

4. Experimental Setup

4.1. Datasets and Federated Client Construction

The testbed combines six publicly sourced text corpora spanning classical English, native-script Tamil, Tanglish, code-mixed text, Tamil-domain reviews, and colloquial English.

The datasets are partitioned into 11 federated clients.

All text is cleaned through whitespace normalization and trimming and tokenized using the DistilGPT2 tokenizer.

The maximum sequence length is 128 tokens with end-of-sequence padding, and causal language-modeling labels are the input token IDs.

Table 2. Personalization scores and adaptive LoRA rank assignment per client.

Client	Train Samples	Score	Rank
client_1	26	0.3250	8
client_2	26	0.3250	8
client_3	27	0.3375	8
client_4	40	0.5000	12
client_5	40	0.5000	12
client_6	40	0.5000	12
client_7	40	0.5000	12
client_8	40	0.5000	12
client_9	40	0.5000	12
client_10	80	1.0000	16
client_11	80	1.0000	16

4.2. Personalization Scores and Adaptive Rank Assignment

Clients 10 and 11 receive the maximum rank of 16, while the lower-volume Shakespeare clients receive rank 8.

4.3. Pilot-Scale Configuration

The results were produced under a quick-validation pilot configuration.

Each of the six source corpora was subsampled to 100 rows before client partitioning.

Local training was capped at 3 optimizer steps, with batch size 2 and learning rate 2×10^{-4} .

Local training sets ranged from 26 to 80 examples and test sets from 7 to 20 examples.

The pilot validates adaptive-rank masking, DP-style clipping and noise, FedProx regularization, and all three CAGER variants end-to-end.

It is not intended to produce production-representative accuracy numbers.

4.4. Training and Privacy Hyperparameters

- Backbone: DistilGPT2 with 82,723,584 total parameters.
- LoRA: maximum rank 16, $\alpha = 32$, dropout = 0.1.
- Adaptive rank: score tiers of 16, 12, 8, and 4.
- FedProx coefficient: $\mu = 0.01$.
- DP-style training: clip norm $C = 1.0$, noise multiplier $z = 0.1$.
- CAGER: 11 experts, router hidden size 256, top- $K = 3$, temperature = 1.0.
- CAGER v2/v3 calibration: 10 samples per client.
- Hardware: CPU-only execution.

5. Results and Discussion

5.1. Overall Model Comparison

The Adaptive PEFT baseline reaches 26.45% Top-1 and 42.68% Top-5 accuracy.

CAGER v1 closely reproduces these values.

CAGER v2 collapses to 0.02% Top-1 accuracy, indicating degenerate routing rather than genuine improvement.

Table 3. Overall model comparison averaged across 11 clients.

Model	T1	T5	PPL	T1G	T5G	PPLG
Adaptive PEFT	26.45	42.68	1019.56	–	–	–
CAGER v1	26.44	42.80	1089.10	-0.01	+0.12	-69.54
CAGER v2	0.02	0.04	1.00	-26.43	-42.64	+1018.56
CAGER v3	26.45	42.80	1083.61	+0.00	+0.12	-64.05

Table 4. Per-client Adaptive PEFT results.

Client	Rank	Base T1	Pers. T1	Gain
client_1	8	9.68	9.68	+0.00
client_2	8	10.77	10.77	+0.00
client_3	8	12.50	12.50	+0.00
client_4	12	56.94	57.03	+0.08
client_5	12	58.44	58.52	+0.08
client_6	12	9.42	9.42	+0.00
client_7	12	5.56	5.56	+0.00
client_8	12	26.93	26.93	+0.00
client_9	12	38.21	38.21	+0.00
client_10	16	58.62	58.62	+0.00
client_11	16	3.68	3.68	+0.00

Personalized CAGER v3 recovers the lost accuracy and improves Top-5 accuracy by 0.12 percentage points.

5.2. Per-Client Results

Accuracy varies substantially across clients, from 3.68% on client_11 to 58.62% on client_10.

The small personalization gains are consistent with the pilot’s extremely limited local exposure of only three optimizer steps.

5.3. CAGER v3 Per-Client Routing Outcomes

Nine of eleven clients show exactly zero change in Top-1 accuracy.

Client 10 shows a +0.12 percentage-point improvement, while client 11 shows a -0.11 percentage-point change.

Perplexity is considerably more volatile, which is attributed primarily to the small number of evaluation tokens in the pilot.

5.4. Qualitative Demonstration

The interactive demonstration automatically detects the most likely client profile and generates a completion.

For the Tamil-script input “ ” (“Today I”), the system identified client_4 (Tamil News) as the dominant profile and routed primarily to client_4, client_9, and client_8 with gate weights 47.0%, 28.0%, and 25.0%, respectively.

Table 5. CAGER v3 per-client Top-1 accuracy and perplexity change.

Client	Adaptive T1	CAGER v3 T1	T1 Gain	PPL Change
client_1	9.68	9.68	+0.00	-0.63
client_2	10.77	10.77	+0.00	-0.09
client_3	12.50	12.50	+0.00	-0.25
client_4	57.03	57.03	+0.00	+0.10
client_5	58.52	58.52	+0.00	-0.13
client_6	9.42	9.42	+0.00	-104.01
client_7	5.56	5.56	+0.00	-253.46
client_8	26.93	26.93	+0.00	-58.57
client_9	38.21	38.21	+0.00	+275.23
client_10	58.74	58.74	+0.12	+0.07
client_11	3.57	3.57	-0.11	-562.79

Table 6. Ablation study over routing components.

Config.	CA	CAIt	EC	OEP	T1	T5	PPL
Adaptive PEFT	Yes	No	No	No	26.45	42.68	1019.56
CAGER v1	Partial	Yes	No	No	26.44	42.80	1089.10
CAGER v2	Yes	Yes	No	No	0.02	0.04	1.00
CAGER v3	Yes	Yes	Yes	Yes	26.45	42.80	1083.61

For the English input “I love”, the system identified client_11 (Colloquial English) and routed to client_11, client_10, and client_3 with gate weights 52.4%, 24.3%, and 23.3%.

The examples show that the router can identify domain-appropriate experts, although the generated completions remain repetitive and may contain malformed sub-word fragments under the pilot training budget.

6. Ablation Study

6.1. Effect of Cross-Attention Routing

Introducing per-client cross-attention routing in CAGER v1 produces a negligible Top-1 change of -0.01 percentage points and a Top-5 improvement of +0.12 percentage points relative to direct adapter use.

This establishes that the routing mechanism is approximately capacity-neutral when trained per client.

6.2. Effect of Removing Client-Specific Router Training

Sharing a single router across all clients in CAGER v2 causes Top-1 accuracy to collapse from 26.44% to 0.02%.

The failure is attributed to degenerate routing caused by the small calibration set and the absence of a mechanism that biases the gate toward the requesting client’s own expert.

6.3. Effect of Expert Compatibility and Own-Expert Prior

CAGER v3 adds client-aware expert compatibility and an own-expert bias, recovering Top-1 accuracy to 26.45% and Top-5 accuracy to 42.80%.

Per-client analysis indicates that the own-expert prior causes the router to select the requesting client’s adapter in most cases, while compatibility scoring allows occasional blending with a related expert.

7. Limitations of the Pilot Evaluation

The results were obtained under a deliberately minimal pilot configuration of 100 samples per source corpus, three optimizer steps per client, and CPU-only execution.

The following limitations should be considered:

- Small per-client datasets and only three optimizer steps limit personalization gains and the statistical reliability of accuracy and perplexity estimates.

- The personalization score currently uses only normalized data volume and does not include linguistic distance or validation-loss signals.
- CAGER v2 and v3 use only 10 calibration samples per client in the pilot, which may be insufficient for stable router learning.
- No formal (ϵ, δ) differential-privacy accounting is reported for the pilot. A rigorous privacy budget requires composition over the complete training schedule using, for example, Rényi-DP analysis.
- CPU-only experiments constrained practical batch size, sequence length, and optimizer steps.
- Fixed-rank LoRA, per-client full fine-tuning, and non-personalized FedAvg baselines have not yet been included.

The immediate next step is to run the identical pipeline at full corpus scale with a realistic training schedule and the missing ablation baselines.

8. Conclusion and Future Work

This paper presented an Adaptive Personalized Federated Learning framework for privacy-preserving multilingual next-word prediction.

The framework combines adaptive-rank LoRA personalization, DP-style gradient clipping and Gaussian noise with Fed-Prox regularization, and a progressive family of Cross-Attention Gated Expert Routers.

On the 11-client, 6-corpus pilot testbed, Adaptive PEFT achieved 26.45% average Top-1 and 42.68% Top-5 accuracy.

Personalized CAGER v3 matched Top-1 accuracy and improved Top-5 accuracy by 0.12 percentage points while recovering from the routing-collapse failure observed in CAGER v2.

Future work will rerun the pipeline at full corpus scale with a realistic training schedule, extend the personalization score using validation loss or embedding distance, report formal differential-privacy accounting, study larger CAGER calibration budgets and anti-collapse regularization, and add fixed-rank LoRA, per-client full fine-tuning, and non-personalized FedAvg baselines.

References

McMahan, B., Moore, E., Ramage, D., Hampson, S. and y Arcas, B. A. 2017. Communication-efficient learning of deep networks from decentralized data. *Proc. 20th Int. Conf. Artificial Intelligence and Statistics (AISTATS)*.

Li, T., Sahu, A. K., Zaheer, M., Sanjabi, M., Talwalkar, A. and Smith, V. 2020. Federated optimization in heterogeneous networks. *Proc. Machine Learning and Systems (MLSys)*.

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L. and Chen, W. 2022. LoRA: Low-rank adaptation of large language models. *Proc. International Conference on Learning Representations (ICLR)*.

Houlsby, N. et al. 2019. Parameter-efficient transfer learning for NLP. *Proc. International Conference on Machine Learning (ICML)*.

Abadi, M. et al. 2016. Deep learning with differential privacy. *Proc. ACM SIGSAC Conference on Computer and Communications Security (CCS)*.

McMahan, H. B., Ramage, D., Talwar, K. and Zhang, L. 2018. Learning differentially private recurrent language models. *Proc. International Conference on Learning Representations (ICLR)*.

Shazeer, N. et al. 2017. Outrageously large neural networks: the sparsely-gated mixture-of-experts layer. *Proc. International Conference on Learning Representations (ICLR)*.

Fedus, W., Zoph, B. and Shazeer, N. 2021. Switch transformers: scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*.

Hard, A. et al. 2018. Federated learning for mobile keyboard prediction. *arXiv preprint arXiv:1811.03604*.

Sanh, V., Debut, L., Chaumond, J. and Wolf, T. 2019. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *NeurIPS Workshop on Energy Efficient Machine Learning and Cognitive Computing*.

Wolf, T. et al. 2020. Transformers: State-of-the-art natural language processing. *Proc. EMNLP: System Demonstrations*.